MX-5 NB turbo run sheet

Calibration: **mx5_nb_me442_v12.mecal** Date: _____ Ambient: _____ °C Fuel: 98 RON, bought _____

Part A in the garage, before burning v12. Part B is everyday driving on the old fuel, **off boost** (MAP under ~100 kPa), until the tank is low. Part C only on **fresh 98** and only on a closed road, track day or dyno. Log every drive (all channels, 10 Hz, as on 7 Oct) and note the log number in the box.

Before you start

☐ Save a copy of the calibration currently in the car (rollback: mx5_nb_me442_v10.mecal or v7).

☐ Open v12 in MEITE and spot-check: Ign. Adv. (Pri 1) at 2000 / 2500 / 3000 / 3500 / 4000 rpm: 83 kPa = 21.25 / 24.25 / 26.75 / 29.5 / 31.75; 97 kPa = 17.5 / 20.5 / 21.0 / 23.75 / 28.0; 111 kPa = 15.0 / 16.75 / 19.0 / 21.25 / 24.5. Idle OL Duty 80 °C and up = 22.0. Oil Press. HRT first point 0.5 V = 0 kPa; EPS: Oil P Limit 60 / 90 / 130 / 160 / 180 / 190 / 200. Boost Settings: Abs. Max Boost = 190, PWM Min / Max Duty = 0 / 80, PWM Solenoid Control = **Normal**.

Part A: garage

A1. Boost leak test

☐ Cap the turbo inlet. Pressurise the intake to about 1 bar (15 psi) through a fitting (e.g. a vacuum port or the BOV line).

☐ Listen, and spray soapy water on: compressor outlet, intercooler and its couplers, throttle body, BOV, vacuum lines, PCV, solenoid hoses.

☐ Leaks found and fixed: _____

A slow leak-down through the engine itself (open valves) is normal. Hissing at a joint is not.

A2. Burn v12, then warm idle check

☐ Burn v12. Warm fully (coolant 85 °C+).

☐ Fan off, wheel straight: idle _____ rpm (expect ~1,000). Fan on: _____ rpm (expect ~1,050–1,100).

☐ Engine off, key on: oil pressure reads ~0 kPa (was ~113 before v12)? Y / N

Log no: _____

Part B: everyday driving on the old fuel (off boost)

B1. Steady cruise for fuelling (VE)

☐ Hold steady speeds for 10 s or more in 3rd, 4th and 5th, spread across 1,500–4,000 rpm, at light and moderate throttle. A few long steady stretches beat lots of short ones.

Log nos: _____ _____ _____

B2. Part-throttle roll-ons (the v8 timing area)

☐ In 3rd from ~2,000 rpm, moderate throttle up to ~3,000 rpm, staying under ~100 kPa. Three times.

☐ Any knock retard (Ign. Adv. Knock Add below 0)? Y / N At rpm / kPa: _____

Log no: _____

B3. Return to idle (baseline, not changed yet)

☐ A few blips in neutral and a few closed-throttle decels to idle. Lowest rpm seen: _____. Any stall? Y / N. Log no: _____

Part C: fresh 98, closed road / track / dyno

C0. Fuel

☐ Run the tank low, then fill with fresh 98. Write the date at the top. Drive ~20 km before Part C so the new fuel is through.

C1. Knock runs

☐ Pull away gently in 1st, three times.

☐ Moderate-throttle roll-on in 3rd from ~1,500 rpm into light boost, three times.

☐ Knock retard seen? Y / N At rpm / kPa: _____

Log no: _____

C2. Spring-only boost (0% duty)

☐ Boost Mode = **Manual Duty**, PWM Manual Duty = **0**.

☐ One full-throttle pull in 3rd from ~2,500 to ~5,500 rpm.

☐ Peak MAP: _____ kPa at _____ rpm. Where it levelled off: _____ kPa.

☐ Afterwards, in the log: Boost Final Duty stayed at 0 for the whole pull? Y / N

Expect about 135–170 kPa (green spring). If MAP goes past ~175 kPa and keeps rising: lift. The 190 kPa over-boost cut is the backstop.
Log no: _____

C3. Manual duty steps

☐ Still in Manual Duty. The same 3rd-gear full-throttle pull at each step. **More duty should give more boost.** If boost goes *down* as duty goes up, stop: the plumbing is still wrong. Stop increasing once the peak reaches ~180 kPa (the cut is at 190).

Duty %	Peak MAP (kPa)	RPM at peak	Knock retard? (Y/N)	AFR at peak	Log no
20					
30					
40					
50					
60					

Afterwards: Boost Mode back to **Closed Loop**. Leave the over-boost cut at 190 until the duty tables are set from these results.

Abort immediately (lift, note the log time) if

- Oil pressure warning, or the 2,000 rpm oil protection limit trips.
- Lean protection trips (3,000 rpm limit), or AFR goes above ~12.5 on boost.
- MAP passes ~175 kPa at 0% duty, overshoots a duty step by more than ~15 kPa, or the over-boost cut fires.
- Knock retard reaches more than ~3° repeatedly in the same area.
- Coolant above ~100 °C or oil above ~120 °C.

After each session

☐ Copy all logs off and note which log matches which test.

☐ Photograph both pages of this sheet; push the logs and photos to the repo.